

Cross-Modal Agreement-Aware Uncertainty-Calibrated Video-IMU Latent World Model for Multi-Horizon Forecasting of Human Motor States

মানব মোটর স্টেটের multi-horizon forecasting-এর জন্য video-IMU ভিত্তিক
uncertainty-calibrated latent world model

Freezing-of-Gait in Parkinson's disease is used as a focused application benchmark. The primary research direction is vision-centric world AI and reliable human-dynamics forecasting.

Vision-centric world AI

Aligned video + IMU

Forecasting, not diagnosis

Existing architectures

Diffusion as future extension

Scope decision

No new backbone architecture will be designed. Established pretrained architectures will be adapted and fine-tuned; the contribution is the agreement-aware forecasting, uncertainty calibration, and evaluation protocol.

নতুন backbone architecture তৈরি করা হবে না। Existing architecture ব্যবহার করে training বা fine-tuning করা হবে; contribution থাকবে agreement-aware forecasting, uncertainty calibration এবং evaluation protocol-এ।

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Supervisor discussion-এর জন্য প্রস্তুত | Proposal version: 1.0 | ২৪ আগস্ট ২০২৬

Project in one page

Core research question

Can temporally aligned video and wearable-IMU models independently forecast the same future motor state, and can their agreement or disagreement provide a calibrated signal of forecast reliability?

মূল প্রশ্ন: একই future motor state-এর forecast-এ video এবং wearable IMU model-এর agreement/disagreement কি prediction reliability-এর calibrated signal দিতে পারে?

01

What

Multi-horizon forecasting of episode-level human motor states from aligned video and IMU.

02

Why

Reliable AI should know when its future forecast may be wrong or uncertain.

03

How

Existing encoders, same-target forecasts, agreement score, latent rollout, calibration.

Final scope boundaries

Item	Decision
Research identity	Vision-centric world AI, latent dynamics, reliable human forecasting.
Application benchmark	Freezing-of-Gait (FOG) in Parkinson's disease.
Forecast target	Operational episode phases: normal, pre-onset, FOG, recovery.
World model claim	Task-specific passive latent world model; not a general-purpose physical simulator.
Architecture policy	No custom backbone. DINOv2, PatchTST, RSSM/Dreamer-style components.
Diffusion	Separate pet-project extension for sampling future latent, pose, or IMU trajectories.

Novelty judgement

FOG detection and generic multimodal fusion are crowded. Video-IMU cross-modal pretraining, latent video world models, and pre-fusion calibration already exist as neighbouring directions. The proposed intersection remains defensible if the paper focuses on same-target multi-horizon forecasting, agreement-aware uncertainty, modality failure, and subject-disjoint/domain-shift evaluation.

Novelty-এর বাস্তব বিচার

FOG detection এবং সাধারণ multimodal fusion crowded। কিন্তু একই future target-এর multi-horizon forecasting, modality agreement দিয়ে uncertainty, missing modality এবং domain shift একসাথে evaluate করা এখনও defensible research gap।

This is a proposal-level literature judgement, not a formal systematic review. A final related-work search should be completed before submission.

Abstract, objectives and hypotheses

Abstract

World models aim to learn compact internal representations of an environment and forecast how its future states evolve. This proposal develops a task-specific, vision-centric latent world model for multi-horizon forecasting of human motor states from temporally aligned video and wearable inertial measurements. Freezing of gait (FOG) in Parkinson's disease is used as an application benchmark and stress test rather than as the primary research domain. Video and IMU streams will be processed by established modality-specific encoders and forecasting models, with both branches predicting the same future motor phase at identical horizons. Their predictive agreement will be evaluated before standard late fusion, while disagreement will provide evidence of possible unreliability. The fused representation will then drive an RSSM-based probabilistic latent dynamics model that generates stochastic multi-step rollouts and calibrated confidence estimates. Operational phases, including normal gait, pre-onset, FOG, and recovery, will be derived from expert-annotated event boundaries and will not be interpreted as longitudinal clinical disease stages. Experiments will compare video-only, IMU-only, naive fusion, and agreement-aware fusion under subject-disjoint splits, missing-modality conditions, and external sensor-domain shift. Evaluation will include future-state accuracy, event-onset lead time, negative log-likelihood, Brier score, calibration error, reliability diagrams, and horizon-dependent uncertainty. The expected contribution is a compute-efficient multimodal latent world-model pipeline for reliable human-dynamics forecasting, with a future extension toward conditional latent diffusion simulation.

Research objectives

1. Construct a temporally aligned video-IMU forecasting benchmark from a public FOG dataset.
2. Fine-tune established video and time-series models to forecast the same future motor phase.
3. Measure cross-modal agreement before fusion and test whether disagreement predicts reliability.
4. Use an existing RSSM-style latent dynamics model for stochastic multi-step rollouts.
5. Evaluate accuracy, early-warning lead time, calibration, missing modality, and distribution shift.

Research hypotheses

- Aligned video-IMU fusion will improve future-state forecasting over either modality alone.
- Cross-modal disagreement will increase under corrupted or unfamiliar conditions.
- Agreement-aware uncertainty will be better calibrated than confidence from a single modality.
- Latent stochastic rollouts will provide useful horizon-dependent forecasts without pixel-level video generation.

সারসংক্ষেপ, উদ্দেশ্য এবং hypothesis

সারসংক্ষেপ

World model এমন একটি compact internal representation শেখে, যার মাধ্যমে environment বা মানুষের future state forecast করা যায়। এই proposal-এ temporally aligned video এবং wearable IMU ব্যবহার করে human motor state-এর multi-horizon forecasting-এর জন্য একটি task-specific, vision-centric latent world-model pipeline তৈরি করা হবে। Parkinson's disease-এর Freezing of Gait (FOG) এখানে মূল research identity নয়; এটি একটি focused application benchmark এবং stress test হিসেবে ব্যবহৃত হবে। Video এবং IMU-এর জন্য established modality-specific model ব্যবহার করা হবে এবং উভয় branch একই future motor phase একই horizon-এ forecast করবে। Standard late fusion-এর আগে দুই prediction-এর agreement মাপা হবে; disagreement সম্ভাব্য unreliability-এর evidence হিসেবে uncertainty estimation-এ ব্যবহার করা হবে। এরপর fused representation একটি existing RSSM-based probabilistic latent dynamics model-এ দেওয়া হবে, যা stochastic multi-step rollout তৈরি করবে। Normal gait, pre-onset, FOG এবং recovery phase expert-annotated event boundary থেকে operational label হিসেবে তৈরি হবে; এগুলো longitudinal clinical disease stage নয়। Video-only, IMU-only, naive fusion এবং agreement-aware fusion subject-disjoint split, missing modality এবং external sensor-domain shift-এর অধীনে তুলনা করা হবে। Accuracy, early-warning lead time, NLL, Brier score, calibration error, reliability diagram এবং horizon-dependent uncertainty দিয়ে evaluation করা হবে। ভবিষ্যতে এই latent dynamics model-কে conditional latent diffusion simulator-এ রূপান্তর করা যাবে।

গবেষণার উদ্দেশ্য

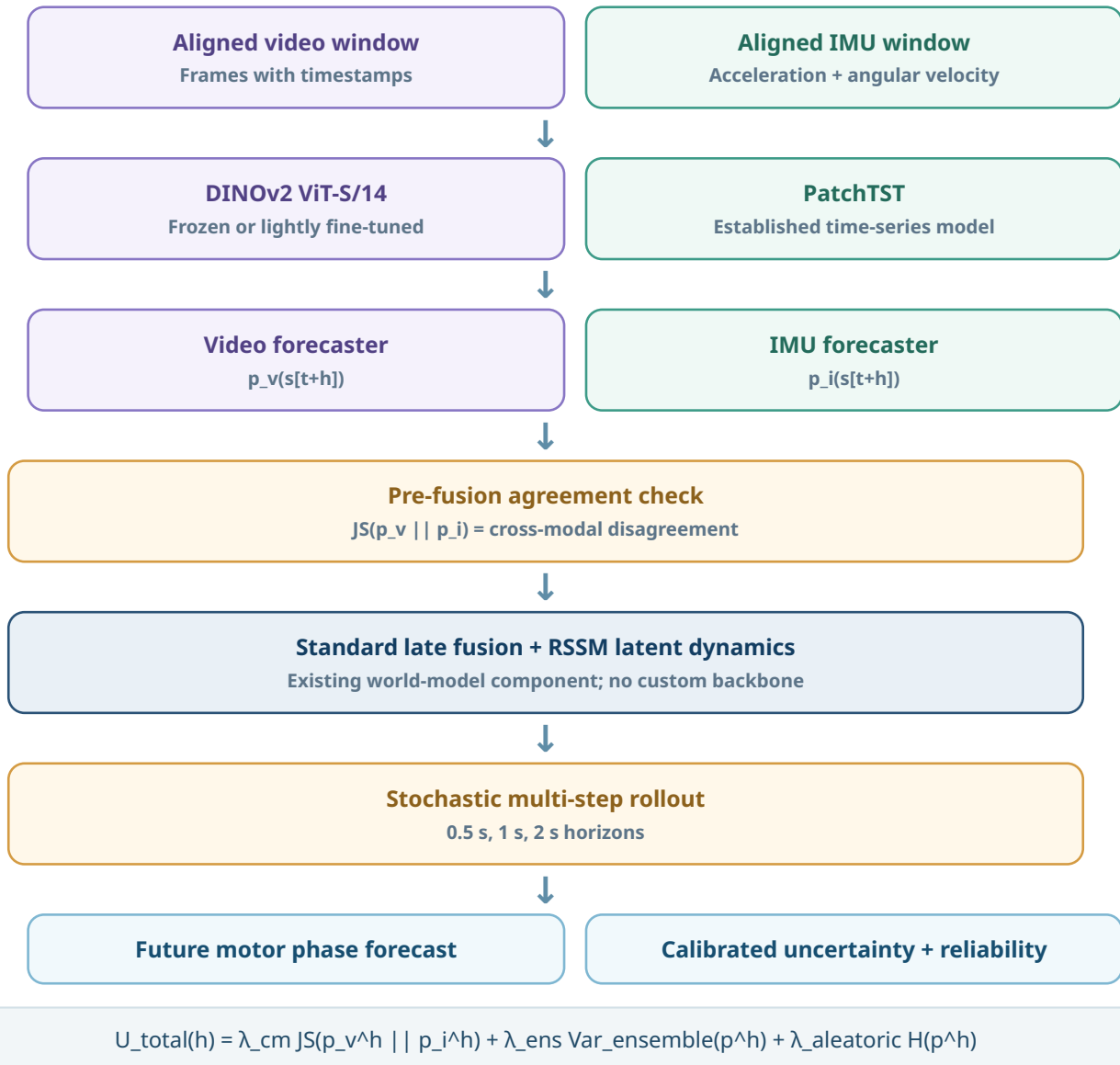
1. Public FOG dataset থেকে temporally aligned video-IMU benchmark তৈরি করা।
2. Established video এবং time-series model fine-tune করে একই future motor phase forecast করা।
3. Fusion-এর আগে cross-modal agreement মাপা এবং disagreement reliability predict করে কি না পরীক্ষা করা।
4. Existing RSSM-style latent dynamics model দিয়ে stochastic multi-step rollout তৈরি করা।
5. Accuracy, early warning, calibration, missing modality এবং domain shift evaluate করা।

গবেষণা hypothesis

- Aligned video-IMU fusion একক modality-এর চেয়ে ভালো future-state forecast দেবে।
- Corrupted বা unfamiliar condition-এ cross-modal disagreement বাড়বে।
- Agreement-aware uncertainty single-modality confidence-এর চেয়ে ভালো calibrated হবে।
- Pixel-level future video না বানিয়েও latent stochastic rollout useful forecast দিতে পারবে।

Research workflow

The central design choice is that both modalities forecast the same future phase before fusion. This is the confirmation step that informs uncertainty.



What makes it a world model?

- It maintains a latent state.
- It models future state transitions.
- It produces multi-step stochastic rollouts.
- It decodes rollouts into future motor states.

কেন এটি world model?

- এটি latent state রাখে।
- Future state transition model করে।
- একাধিক সম্ভাব্য future rollout তৈরি করে।
- Rollout-কে future motor state-এ decode করে।

Models, training and uncertainty

Established model components

Component	Selected existing model	Role and training
Visual encoder	DINOv2 ViT-S/14	Extract frame-level semantic features. Freeze initially; optionally fine-tune the final blocks with a low learning rate.
Video temporal model	Standard Transformer Encoder	Aggregate cached visual embeddings over the observation window and forecast future motor phase.
IMU model	PatchTST	Model acceleration and angular-velocity sequences. Train on the aligned task windows.
World model	RSSM from Dreamer-style models	Learn probabilistic latent transitions and multi-step future rollouts. No actor-critic or control claim in the FYDP.
Fusion	Standard late probability fusion	Combine calibrated modality predictions; no custom fusion backbone.

Training protocol

1. Synchronize video frames and IMU timestamps; create fixed observation windows.
2. Construct the same target label and horizon for both branches.
3. Train video-only and IMU-only forecasters.
4. Calibrate each branch on a validation set, then compute agreement/disagreement.
5. Train or fine-tune the RSSM using fused context and future encoded states.
6. Use 3-5 different random seeds as a deep ensemble.

Important evaluation rule: latent rollout consistency may be reported as a diagnostic, but the main performance measure must be decoded future motor-state accuracy because a learned latent state has no independent ground-truth label.

Uncertainty evaluation

Uncertainty source	Operational measure
Cross-modal disagreement	Jensen-Shannon divergence between calibrated video and IMU distributions.
Aleatoric uncertainty	Predictive entropy or heteroscedastic transition variance.
Epistemic uncertainty	Variance across 3-5 independently trained model seeds.
Calibration	ECE, Brier score, NLL, reliability diagrams, and interval/coverage analysis.

Model, training এবং uncertainty

ব্যবহৃত existing model

অংশ	Model	কাজ এবং training
Visual encoder	DINOv2 ViT-S/14	প্রথমে frozen থাকবে। প্রয়োজন হলে শেষের block-গুলো low learning rate-এ fine-tune করা হবে।
Video temporal model	Standard Transformer Encoder	Cached visual embedding-এর sequence থেকে temporal pattern এবং future phase forecast করবে।
IMU model	PatchTST	Acceleration এবং angular velocity sequence model করবে।
World model	Dreamer-style RSSM	Probabilistic latent transition এবং multi-step future rollout করবে। FYDP-তে control বা actor-critic claim থাকবে না।
Fusion	Standard late probability fusion	Calibrated video এবং IMU prediction combine করবে; custom fusion backbone নয়।

Training protocol

1. Video frame এবং IMU timestamp synchronize করে fixed observation window তৈরি করা।
2. দুই branch-এর জন্য একই target label এবং একই horizon তৈরি করা।
3. Video-only এবং IMU-only forecaster train করা।
4. Validation set-এ branch calibration করে agreement/disagreement মাপা।
5. Fused context দিয়ে RSSM train বা fine-tune করা।
6. ৩-৫টি random seed-এর deep ensemble ব্যবহার করা।

গুরুত্বপূর্ণ: Latent rollout consistency diagnostic হিসেবে দেখানো যাবে, কিন্তু main performance হবে decoded future motor-state accuracy। কারণ learned latent state-এর আলাদা ground truth থাকে না।

Uncertainty evaluation

Uncertainty source	পরিমাপ
Cross-modal disagreement	Calibrated video এবং IMU distribution-এর Jensen-Shannon divergence।
Aleatoric uncertainty	Predictive entropy অথবা transition variance।
Epistemic uncertainty	৩-৫টি independently trained seed-এর prediction variance।
Calibration	ECE, Brier score, NLL, reliability diagram এবং coverage analysis।

Dataset, labels and evaluation

Main multimodal dataset

A Public Dataset of Videos, Acceleration, Angular Velocity, and Clinical Scales in Individuals with Parkinson's Disease During the Turning-in-Place Task contains video, wearable acceleration/angular velocity, specialist-reviewed freezing events, and metadata for 35 participants. The Figshare page reports an all-download size of approximately 4.82 GB. This is the primary video-IMU dataset; the full 70 GB tdcsfog/defog download is not required for the main experiment.

External validation

FoG-STAR provides a sensor-only external benchmark with 22 participants, four IMUs, and 101 annotated FOG episodes. Unless public video access is confirmed, it should be used for sensor-domain robustness rather than sample-level video-IMU fusion. tdcsfog-to-defog may be an optional sensor-only lab-to-home shift experiment.

Label and split design

Design element	Decision
Motor phases	Normal, pre-onset window, FOG episode, recovery; derived from expert event boundaries.
Forecast horizons	Example: 0.5 s, 1 s, and 2 s. Final values depend on sampling and annotation resolution.
Data split	Subject-disjoint group split or grouped cross-validation. No overlapping windows across train/test.
Class imbalance	Weighted loss or focal loss; report AUPRC because FOG events are sparse.
Cross-dataset rule	Do not pair video from one dataset with IMU from another. Use external datasets only for transfer/robustness.

Metrics

- Macro F1 and balanced accuracy
- AUPRC for sparse FOG events
- Event-onset lead time
- Future-state accuracy at each horizon
- Negative log-likelihood
- Brier score and ECE
- Reliability diagrams
- Uncertainty under shift and missing modality

Clinical wording: The study forecasts episode-level motor states. It does not predict the next clinical stage of Parkinson's disease, which would require longitudinal repeated clinical assessments.

Dataset, label এবং evaluation

Main multimodal dataset

A Public Dataset of Videos, Acceleration, Angular Velocity, and Clinical Scales in Individuals with Parkinson's Disease During the Turning-in-Place Task dataset-এ ৩৫ জন participant-এর video, wearable acceleration/angular velocity, specialist-reviewed freezing event এবং metadata আছে। Figshare অনুযায়ী all-download size প্রায় ৪.৮২ GB। এটিই main video-IMU dataset; ৭০ GB-এর tdcsfog/defog full download main experiment-এর জন্য দরকার নেই।

External validation

FoG-STAR dataset-এ ২২ জন participant, ৪টি IMU এবং ১০১টি annotated FOG episode আছে। Public video access নিশ্চিত না হলে এটিকে sensor-only external validation হিসেবে ব্যবহার করতে হবে। tdcsfog-to-defog optional lab-to-home sensor shift experiment হতে পারে।

Label এবং split

বিষয়	সিদ্ধান্ত
Motor phase	Normal, pre-onset, FOG episode এবং recovery; expert event boundary থেকে তৈরি operational label।
Forecast horizon	উদাহরণ: ০.৫ সেকেন্ড, ১ সেকেন্ড এবং ২ সেকেন্ড।
Data split	Subject-disjoint group split; train/test-এ overlapping window থাকবে না।
Class imbalance	Weighted বা focal loss; sparse FOG event-এর জন্য AUPRC report করা হবে।
Cross-dataset rule	এক dataset-এর video এবং অন্য dataset-এর IMU sample-wise pair করা যাবে না।

মূল evaluation metrics

Macro F1, balanced accuracy, FOG-এর জন্য AUPRC, event-onset lead time, future-state accuracy, NLL, Brier score, ECE, reliability diagram এবং missing-modality/domain-shift uncertainty report করা হবে।

Clinical wording: এই study episode-level motor state forecast করবে। Parkinson's disease-এর next clinical stage বা disease progression forecast করবে না।

Technical stack, timeline, compute and publication

Technical stack

Python, PyTorch, CUDA, DINOv2, PatchTST, an RSSM implementation, OpenCV/FFmpeg/Decord, NumPy, SciPy, pandas, scikit-learn, TorchMetrics, custom calibration metrics, Hydra, Weights & Biases or MLflow, Git, and cached embeddings stored in Parquet/HDF5/Zarr.

24-week timeline

Weeks	Milestone
1-2	Literature review, dataset/license audit, final protocol.
3-5	Timestamp alignment, phase-label construction, subject-disjoint split.
6-8	Video-only and IMU-only baselines.
9-11	Same-target multi-horizon forecasting.
12-14	Agreement measurement, calibration, standard late fusion.
15-17	RSSM-based probabilistic latent rollout.
18-21	Deep ensemble, missing modality, external sensor shift.
22-24	Ablations, statistics, paper, thesis, code release.

Compute estimate

One RTX 3060 12 GB is workable when DINOv2 is frozen and visual embeddings are cached. A 16 GB GPU is more comfortable. Recommended system resources are 32 GB RAM and 150-300 GB SSD. Full V-JEPA-style pretraining or raw video diffusion is outside the FYDP compute scope.

Publication and future diffusion

Publication

Publishable as a methods-and-application paper if it includes strong unimodal baselines, leakage-free subject splits, calibration analysis, missing-modality tests, external validation, and reproducible code. IJABC is a realistic target because its official J-STAGE page describes it as peer-reviewed, open access, and within medicine, engineering, information sciences, and psychology/education.

Diffusion pet project

After the FYDP, condition a latent diffusion model on the learned world state and generate multiple future latent, pose, or IMU trajectories. Start with latent/pose simulation; raw RGB video generation is a separate, much larger project.

Supervisor decision requested: approve the focused FYDP as a task-specific latent world-model forecasting study, with diffusion simulation listed as a future extension rather than a core deliverable.

Technical stack, timeline, compute এবং publication

Technical stack

Python, PyTorch, CUDA, DINOv2, PatchTST, RSSM implementation, OpenCV/FFmpeg/Decord, NumPy, SciPy, pandas, scikit-learn, TorchMetrics, calibration metrics, Hydra, Weights & Biases বা MLflow, Git এবং cached embedding-এর জন্য Parquet/HDF5/Zarr ব্যবহার করা হবে।

২৪ সপ্তাহের timeline

সময়	কাজ
১-২	Literature review, dataset/license audit, final protocol।
৩-৫	Timestamp alignment, phase label এবং subject-disjoint split।
৬-৮	Video-only এবং IMU-only baseline।
৯-১১	Same-target multi-horizon forecasting।
১২-১৪	Agreement measurement, calibration এবং late fusion।
১৫-১৭	RSSM-based probabilistic latent rollout।
১৮-২১	Deep ensemble, missing modality এবং external sensor shift।
২২-২৪	Ablation, statistics, paper, thesis এবং code release।

Computational power

একটি RTX 3060 12 GB যথেষ্ট workable, যদি DINOv2 frozen রেখে visual embedding cache করা হয়। ১৬ GB GPU আরও comfortable। ৩২ GB RAM এবং ১৫০-৩০০ GB SSD ভালো হবে। Full V-JEPA-style pretraining বা raw video diffusion FYDP-এর scope নয়।

Publication এবং diffusion extension

Publication

Strong baseline, leakage-free split, calibration, missing-modality, external validation এবং reproducible code থাকলে methods-and-application paper হিসেবে publish করা সম্ভব। IJABC-এর official scope-এর সঙ্গে এটি মানানসই।

Diffusion pet project

FYDP-এর পরে learned world state condition করে latent diffusion model দিয়ে multiple future latent, pose বা IMU trajectory generate করা যাবে। Raw RGB video দিয়ে শুরু করা উচিত নয়।

Supervisor-এর কাছে সিদ্ধান্ত: FYDP-কে task-specific latent world-model forecasting study হিসেবে approve করা; diffusion simulation-কে future extension রাখা।

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Supervisor discussion checklist

- Approve the title and the FOG-as-application framing.
- Approve the 4.82 GB synchronized dataset as the primary data source.
- Approve existing-model adaptation instead of new architecture design.
- Approve subject-disjoint splits and calibration as mandatory evaluation.
- Keep diffusion simulation outside the core FYDP deliverable.

 Student

 Supervisor